

① Linear Equation -

$$(1) \frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$$

$$(2) \frac{dy}{dx} (x^2+1) + 4xy = x$$

$$(3) (\cos^2 x - y \cos x) dx - (1 + \sin x) dy = 0$$

$$(4) (y \sin 2x - \cos x) dx + (1 + \sin^2 x) dy = 0$$

$$(5) e^x [y - 3(e^x + 1)^2] dx + (e^x + 1) dy = 0, \quad y(0) = 4$$

$$(6) \frac{dr}{d\theta} + r \tan \theta = \cos^2 \theta, \quad r(\pi/4) = 1$$

$$(7) 2x(y+1)dx - (x^2+1)dy = 0, \quad y(1) = -5$$

② Bernoulli's Equation -

$$(1) \frac{dy}{dx} + y = xy^3$$

$$(2) \frac{dy}{dx} - \frac{y}{x} = -\frac{y^2}{x}$$

$$(3) x \frac{dy}{dx} + y = -2x^6 y^4$$

$$(4) dy + (4y - 8y^3)x dx = 0$$

$$(5) \frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}, \quad y(1) = 2$$

$$(6) x \frac{dy}{dx} + y = (xy)^{3/2}, \quad y(1) = 4$$

③ Homogeneous Linear system of equations -

✓ (1) $x' = 6x - 3y$

$$y' = 2x + y$$

(2) $\frac{dx}{dt} = 5x - 2y$

$$\frac{dy}{dt} = 4x - y$$

✓ (3) $\frac{dx}{dt} = -2x + 7y$

$$x(0) = 9, y(0) = -1$$

$$\frac{dy}{dt} = 3x + 2y$$

④ Higher order first degree Linear Differential Equations -

(1) $y^{iv} - 3y''' + 2y'' = 0$

(2) $y''' + y'' + 3y' - 5y = 0$

✓ (3) $y^{iv} - y = 0$

(4) $y''' - 4y'' + y' + 6y = 0$

✓ (5) $y'' - 4y' + 3y = 9x^2 + 4$ $y(0) = 6, y'(0) = 3$

✓ (6) $y'' - 4y' + 13y = 8 \sin 3x$ $y(0) = 1, y'(0) = 2$

⑤

(7) $(2D^2x - 5DxDy + 2D^2y)z = 24(y-x)$

⑤ Higher order non-homogeneous LDE with constant coefficients - (UC method)

(1) $y'' - 2y' - 3y = 2e^x - 10 \sin x$, $y(0) = 2$, $y'(0) = 4$

(2) $y'' - 4y' + 3y = 9x^2 + 4$, $y(0) = 6$, $y'(0) = 3$

(3) $y'' - 4y' + 13y = 8 \sin 3x$

(4) $y'' - 2y' - 8y = 4e^{2x} - 21e^{3x}$

(5) $y'' + 4y' + 13y = 18e^{2x}$

⑥ Cauchy-Euler Equation -

(1) $x^2 y'' + xy' + 4y = 2x \ln x$

(2) $x^2 y'' - 4xy' + 4y = 4x^2 - 6x^3$, $y(2) = 4$, $y'(2) = -1$

(3) $x^3 y''' - x^2 y'' + 2xy' - 2y = x^3$

(4) $x^2 y'' + xy' + y = 4 \sin(\ln x)$

(5) $x^3 \frac{d^3 y}{dx^3} - 4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} = 4 \ln x$

⑦ Linear Partial Differential Equation of Order One -

(1) $z = (ax + \frac{y}{a} + b)^2$

(2) $z = Ae^{Pt} \sin P_0 x$

(3) $z = a(x+y) + b(x-y) + abt$

(4) $z = axe^y + \frac{1}{2} a^2 e^{2y} + b$

(5) $z = y^2 + 2t \left(\frac{1}{x} + \ln y \right)$

$$(6) z = f(x+iy) + F(x-iy), (i^2 = -1)$$

$$(7) z = f(x-at) + F(x+at)$$

$$(8) z = f(xy) + g(x/y)$$

⑧ Linear Partial Differential Equation of Order One - (Lagrange's Method)

$$(1) y^2 p - xyq = xz$$

$$(2) p + 3q = 5z + \tan(y-3x)$$

$$(3) p - 2q = 3x^2 \sin(y+2x)$$

$$(4) y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} = xyz^2 (x^2 - y^2)$$

$$(5) xy \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = zxy - 2x^2$$

$$(6) (mz - ny) \frac{\partial z}{\partial x} + (nx - lz) \frac{\partial z}{\partial y} = \ln y - mx$$

$$(7) x(y^2 - z^2)p - y(x^2 + z^2)q = z(x^2 + y^2)$$

$$(8) z(x+y) \frac{\partial z}{\partial x} + z(x-y) \frac{\partial z}{\partial y} = x^2 + y^2$$

⑨ Linear Partial Differential Equations of higher order -

$$(1) (D^2x - 2DxDy + D^2y)z = e^{3x-2y}$$

$$(2) (D^3x - 4D^2xDy + 4DxD^2y)z = \cos(x+3y)$$

$$(3) (D^2x - 5DxDy + 4D^2y)z = \sin(4x+2y)$$

$$(4) x + y - 2z = (2x + y)^{1/2}$$

$$(5) (D^3x - 6D^2xDy + 11DxD^2y - 6D^3y)z = e^{5x+6y}$$

$$(6) (D^3x - 4D^2xDy + 4DxD^2y)z = 6 \sin(3x+2y)$$